

MITHESH R MOHAN

Aspiring Software Engineer | Computer Science & Engineering

+91 9074073721 | mitheshrmohan05@gmail.com | linkedin.com/in/mithesh-r-mohan | github.com/Mithesh-R-Mohan

PROFESSIONAL SUMMARY

Detail-oriented Computer Science undergraduate specializing in Cyber Security and Digital Forensics with a strong grounding in Data Structures & Algorithms, Object-Oriented Programming, and systems development. Experienced in engineering security monitoring tools using Python and C++, with hands-on implementation of machine learning classification pipelines and real-time network traffic analysis. Eager to contribute to scalable enterprise software development and defensive system architecture.

TECHNICAL SKILLS

- **Languages & Databases:** Python, C++, SQL, SQLite, MySQL
- **Frameworks & Libraries:** Scikit-learn, Tkinter (GUI), HTML5, CSS3
- **Tools & Environment:** Git, GitHub, Linux/Unix Shell, Wireshark, Network Packet Inspection
- **Core Competencies:** Data Structures & Algorithms, OOPs, Network Security, Machine Learning Fundamentals

TECHNICAL PROJECTS

Real-Time Phishing URL Detection System (Edge Monitor)

Python, Scikit-learn, Tkinter, SQLite

Software & Security Developer

- Engineered a real-time desktop security monitor querying the local Microsoft Edge SQLite history database to intercept visited URLs with zero user interruption.
- Built an automated feature extraction engine extracting lexical indicators including URL length, digit/symbol counts, HTTPS validation, and targeted phishing tokens.
- Trained and evaluated a Random Forest Classifier on labeled datasets to accurately classify live URLs as Legitimate or Phishing.
- Designed an interactive desktop GUI using Tkinter to visualize live detection logs, classification outcomes, and real-time model evaluation metrics.

Network Intrusion Detection & Traffic Analysis System

Python, Linux CLI, Socket Programming

Systems & Network Developer

- Developed a Python-based packet analysis engine capable of processing live network packet streams with zero buffer drop during simulated test workloads.
- Programmed signature and anomaly detection routines covering common protocol vulnerabilities (TCP SYN floods, abnormal port sweeps), reducing false-positive alerts by **22%**.
- Formatted continuous event logs into structured tabular summaries, enabling quick triage and root-cause analysis of localized traffic spikes.
- Utilized Linux networking utilities to automate packet capture scheduling and local socket diagnostics.

EDUCATION

Bachelor of Technology in Computer Science & Engineering

08/2023 – 06/2027

VIT Bhopal University | Specialization in Cyber Security and Digital Forensics

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Computer Networks, Operating Systems, Software Engineering.

Higher Secondary Education (12th Standard - PCM)

05/2022 – 03/2023

Lakshmi Narayana Vidyanikethan

CERTIFICATIONS & TRAINING

- **Mastering Data Structures and Algorithms using Python** – Udemy
- **AWS Technical Essentials** – AWS
- **Official Practice Question Set: AWS Certified Cloud Practitioner (CLF-C02)** – AWS
- **The Bits and Bytes of Computer Networking** – Google Coursera
- **Introduction to C++** – Simplilearn